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LAB TASK 04

REGN NO: FA25-BCS-017
DATA STRUCTURE LAB

Lab Task 1

Write a function that deletes all those nodes from a linked list which have even/odd numbered value in their data part.

CODE:

```
#include <iostream>
using namespace std;
```

```
struct Node {
    int data;
    Node* next;
};
```

```
void deleteEvenOdd(Node*& head, int choice) {
```

```
    Node* current = head;
    Node* prev = NULL;
```

```
    while (current != NULL) {
```

```
        if ((choice == 1 && current->data % 2 == 0) ||
            (choice == 2 && current->data % 2 != 0)) {
```

```
            Node* temp = current;
```

```
            if (prev == NULL)
                head = current->next;
```

```
            else
                prev->next = current->next;
```

```
            current = current->next;
            delete temp;
```

```
        }
```

```
        else {
            prev = current;
            current = current->next;
```

```
        }
```

```
    }
```

```
}
```

```

void display(Node* head) {
    while (head != NULL) {
        cout << head->data << " ";
        head = head->next;
    }
}

int main() {

    Node* head = new Node{10, NULL};
    head->next = new Node{15, NULL};
    head->next->next = new Node{20, NULL};
    head->next->next->next = new Node{25, NULL};
    head->next->next->next->next = new Node{30, NULL};

    cout << "Original List: ";
    display(head);

    int choice;
    cout << "\n1. Delete Even Values";
    cout << "\n2. Delete Odd Values";
    cout << "\nEnter choice: ";
    cin >> choice;

    deleteEvenOdd(head, choice);

    cout << "Updated List: ";
    display(head);

    return 0;
}

```

OUTPUT

```
C:\Users\HP\OneDrive\Desktop  X  +  v
Original List: 10 15 20 25 30
1. Delete Even Values
2. Delete Odd Values
Enter choice: 1
Updated List: 15 25
-----
Process exited after 23.94 seconds with return value 0
Press any key to continue . . . |
```

Lab Task 2

Write a function that implements Josephus problem.

CODE:

```
#include <iostream>
using namespace std;

struct Node {
    int data;
    Node* next;
};

int josephus(int n, int k) {

    Node* head = new Node{1, NULL};
    Node* tail = head;

    for (int i = 2; i <= n; i++) {
        tail->next = new Node{i, NULL};
        tail = tail->next;
    }

    tail->next = head;

    Node* current = head;
    Node* prev = tail;

    while (current->next != current) {

        for (int i = 1; i < k; i++) {
            prev = current;
```

```

        current = current->next;
    }

    prev->next = current->next;

    delete current;

    current = prev->next;
}

int winner = current->data;

delete current;

return winner;
}

int main() {

    int n, k;

    cout << "Enter number of people: ";
    cin >> n;

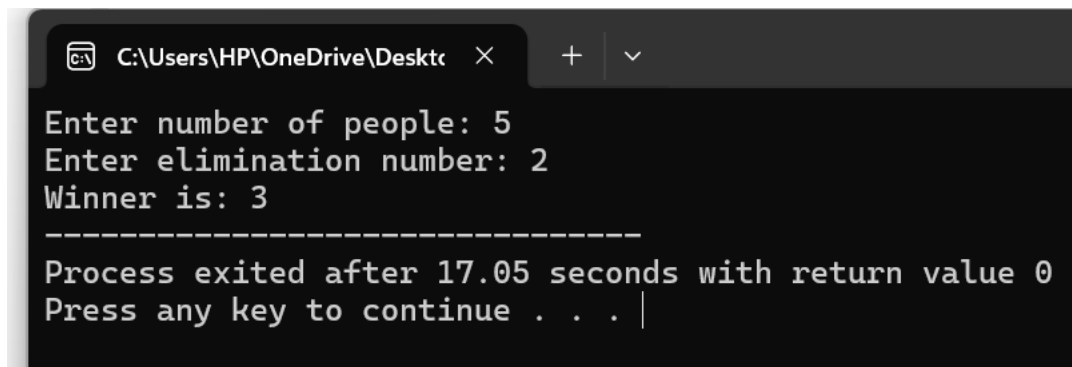
    cout << "Enter elimination number: ";
    cin >> k;

    cout << "Winner is: " << josephus(n, k);

    return 0;
}

```

OUTPUT:



```

C:\Users\HP\OneDrive\Desktop >
Enter number of people: 5
Enter elimination number: 2
Winner is: 3
-----
Process exited after 17.05 seconds with return value 0
Press any key to continue . . . |

```

Lab Task 3

Write a function that deletes all even positioned nodes from a linked list. Last node should also be deleted if its position is even

CODE:

```
#include <iostream>
```

```
using namespace std;
```

```
// Node structure
```

```
struct Node {
```

```
    int data;
```

```
    Node* next;
```

```
};
```

```
// Function to delete even positioned nodes
```

```
void deleteEvenPosition(Node*& head) {
```

```
    Node* current = head;
```

```
    Node* prev = NULL;
```

```
// Position starts from 1
```

```
int position = 1;
```

```
// Traverse linked list
```

```
while (current != NULL) {
```

```
    // Check whether position is even
```

```
    if (position % 2 == 0) {
```

```

// Store node to be deleted
Node* temp = current;

// Connect previous node to next node
prev->next = current->next;

// Move current to next node
current = current->next;

// Delete even positioned node
delete temp;
}
else {
    // Keep odd positioned node
    prev = current;
    current = current->next;
}

// Move to next position
position++;
}
}

// Function to display linked list
void display(Node* head) {

```

```

// Traverse and print all nodes
while (head != NULL) {
    cout << head->data << " ";
    head = head->next;
}
}

int main() {

    // Create linked list
    Node* head = new Node{10, NULL};
    head->next = new Node{20, NULL};
    head->next->next = new Node{30, NULL};
    head->next->next->next = new Node{40, NULL};
    head->next->next->next->next = new Node{50, NULL};
    head->next->next->next->next->next = new Node{60, NULL};

    cout << "Original List: ";
    display(head);

    // Call deletion function
    deleteEvenPosition(head);

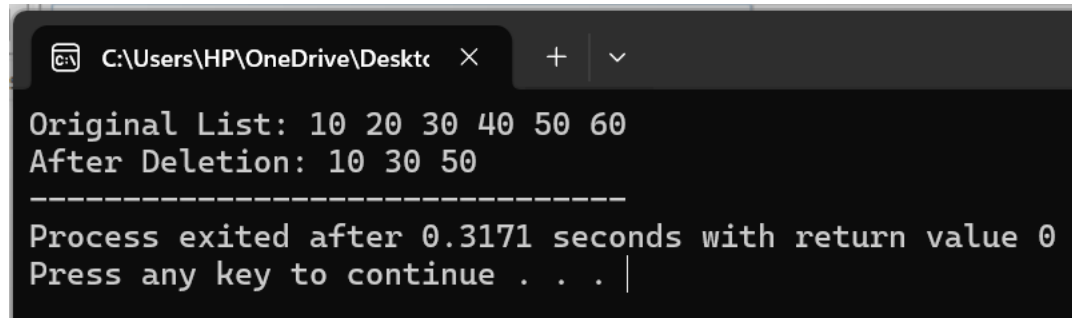
    cout << "\nAfter Deletion: ";
    display(head);

    return 0;
}

```

}

OUTPUT:



A screenshot of a Windows command prompt window. The title bar shows the file path "C:\Users\HP\OneDrive\Desktop" and standard window controls. The command prompt displays the following text: "Original List: 10 20 30 40 50 60", "After Deletion: 10 30 50", a separator line of dashes, "Process exited after 0.3171 seconds with return value 0", and "Press any key to continue . . . |".

```
C:\Users\HP\OneDrive\Desktop >  
Original List: 10 20 30 40 50 60  
After Deletion: 10 30 50  
-----  
Process exited after 0.3171 seconds with return value 0  
Press any key to continue . . . |
```